# MTH2008

## UNIVERSITY OF EXETER

## COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

## MATHEMATICS

## January 2022

### Real Analysis

### Module Leader: Professor Nigel Byott

### Duration: 2 HOURS

**Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).**

*Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.*

This is a **CLOSED NOTE** examination.

No word count

## SECTION A

1. (a) Let $S = (0, 1] \cup \{2\} \cup [3, 4)$.

   (i) Is $S$ open, closed, neither, or both? Justify your answer.

   (ii) Find the complement $S^c$ of $S$ in $\mathbb{R}$.

   (iii) Find the boundary $\partial S$ of $S$.

   (iv) Find the set of limit points of $S$. (10)

   (b) Using the $\epsilon$–$\delta$ definitions of limits and continuity, show that the function $f(x) = x^2 - x\sin(x)$ is continuous at $x_0 = 0$. (10)

   (c) Let $(a_n)_{n\geq 1}$ be a sequence of real numbers such that $a_n \to \infty$ as $n \to \infty$, and let $(b_n)_{n\geq 1}$ be an arbitrary sequence of real numbers. Prove that $a_n^2 + b_n^2 \to \infty$ as $n \to \infty$. (10)

   (d) Show that the function $f : [0, 1] \to \mathbb{R}$ given by $f(x) = 1 - x$ is Riemann integrable, and evaluate $\int_0^1 f(x)\,\mathrm{d}x$ using Riemann sums. (10)

   (e) Find the derivative of the function $f : \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$f\big((x_1, x_2)^T\big) = \big(\sin(x_1)\cos(x_2), \cos(x_1)\sin(x_2)\big)^T.$$

   (10)

   [**50**]

## SECTION B

2. (a) Let $A$ and $B$ be non-empty, bounded sets of real numbers, let $\alpha = \sup(A)$ and $\beta = \sup(B)$, and suppose that $\alpha < \beta$.

Determine whether each of the following statements is necessarily true, or can be false. Justify your answers with a proof or a counterexample.

(i) $a \leq b$ for all $a \in A$ and $b \in B$;

(ii) there exists an element $b \in B$ such that $b \notin A$;

(iii) if $\beta \notin B$ then $B$ contains infinitely many distinct elements which are upper bounds for $A$. (15)

(b) For $n \geq 2$, let $f_n : [0,1] \to \mathbb{R}$ be the function given by

$$f_n(x) = \begin{cases} 1 - nx & \text{if } 0 \leq x \leq \dfrac{1}{n}, \\ \dfrac{nx-1}{n-1} & \text{if } \dfrac{1}{n} < x \leq 1. \end{cases}$$

Find the pointwise limit $f$ of the sequence of functions $(f_n)_{n\geq 2}$. Does $f_n$ converge to $f$ uniformly on $[0,1]$? Justify your answer. (10)

**[25]**

3. (a) Let $f : [a,b] \to \mathbb{R}$ be a continuous function such that $f(a) > 0$ and $f(b) < 0$, and let

$$S = \{x \in [a,b] : f(x) < 0\}.$$

Show that the set $S$ has an infimum $c$, and prove that $f(c) = 0$. (12)

(b) Let $g : [a,b] \to \mathbb{R}$ be a bounded function, and let $g^+ : [a,b] \to \mathbb{R}$ be defined by

$$g^+(x) = \begin{cases} g(x) & \text{if } g(x) \geq 0, \\ 0 & \text{if } g(x) < 0. \end{cases}$$

(i) Show that, for any closed interval $[c,d] \subseteq [a,b]$, we have

$$0 \leq \sup_{x\in[c,d]} g^+(x) - \inf_{x\in[c,d]} g^+(x) \leq \sup_{x\in[c,d]} g(x) - \inf_{x\in[c,d]} g(x).$$

(ii) Deduce that if $g$ is Riemann integrable on $[a,b]$ then $g^+$ is also Riemann integrable on $[a,b]$. (13)

**[25]**

4. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined at $x = (x_1, x_2)^T$ by

$$f((x_1, x_2)^T) = \begin{cases} \dfrac{2x_1^2 + x_2^2}{x_1^2 + 2x_2^2} & \text{if } x \neq (0,0)^T, \\ 1 & \text{if } x = (0,0)^T. \end{cases}$$

(a) Determine whether $f$ is continuous at $(0,0)^T$. (5)

(b) Show that, for each point $(p_1, p_2)^T \neq (0,0)^T$, $f$ is differentiable at $(p_1, p_2)^T$, and find its derivative there. (10)

(c) Find all the stationary points of $f$. (5)

(d) Does $f$ have a global minimum? Justify your answer. (5)

[**25**]